Pine Beetle Outbreak Analysis and Predictive Solutions

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DS-399: Environmental Data Science
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May 7th 2026

01. Executive Summary: (Suggested to be written last)

- Super briefly introduce pine beetles
- Pine beetles pose a huge threat to forest health!!
- Lots of research and literature but difficult solutions to apply at a large scale
- Problem Overview
    - Lots of damage to trees and forests (majority of mass tree mortality attributed to beetles)
    - Cannot use beetle infested wood in lumber production
    - Wild fire risk
- Findings demonstrate that pine beetles like mild winters and droughts
- Solutions overview
    - Tree thinning
    - Pheromones
    - Pesticides
- Recommendation overview
    - Use a combination of tree thinning, and pheromones/trap trees to mitigate beetles. Use pesticides at a small scale in dry places where needed.
- Remember to keep this section short

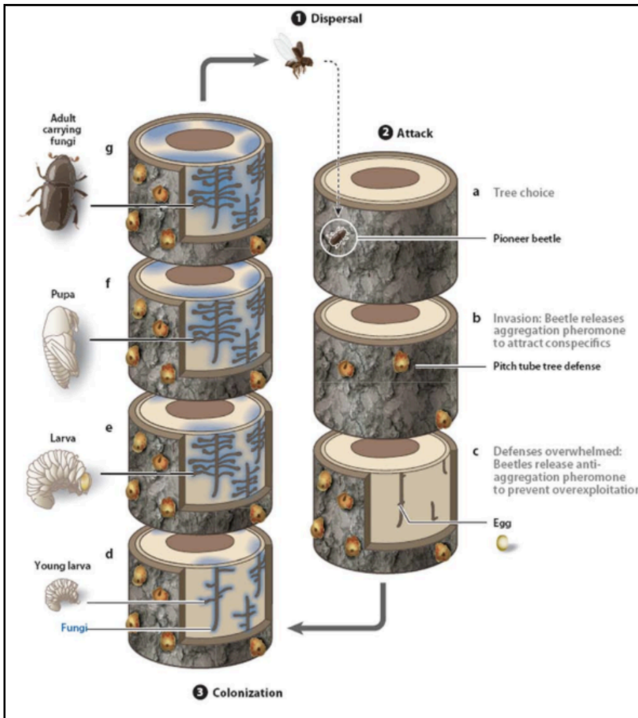
02. Background (needs citations)

The mountain pine beetle (*Dendroctonus ponderosae*) has become one of the most destructive pests in western North America. From the late 1990s to 2013, mountain pine beetles affected 3.4 million acres of forests in Colorado alone, killing up to 90% of trees in some forests. Pine beetles tend to focus on attacking already weakened or stressed trees, but during an outbreak, they can overwhelm healthy trees. They attack most pine species, including ponderosa, lodgepole, whitebark, and limber pine trees. Pine beetles typically have a one-year life cycle, although this can be extended in colder climates. Throughout their life, they go through four stages: egg, larva, pupa, and adult. Most of a pine beetle's lifespan is spent under tree bark, until they emerge in mid-summer (July-August). Upon emerging, they find a new tree to lay eggs in, as they can only live for a few weeks and must quickly locate a new host tree. Once a suitable host is located, they find a mate, lay their eggs, and die, leaving their offspring to repeat the cycle. While it sounds like pine beetles are extremely damaging, in moderation, they are beneficial to overall forest health. At low levels, they help recycle nutrients and clear out old trees, making room for new ones.

Mountain pine beetle (Dendroctonus ponderosae) (Source: Sustainable Forestry Initiative, 2023)

Their issue comes when their populations grow out of balance, often due to warmer temperatures, especially in the winter.

There are many factors that contribute to the severity of an outbreak, but the primary variables are climate, tree stress, and how beetles communicate with each other. Generally, extreme cold kills large portions of beetle populations, which would keep pine beetle populations in moderation. However, as winters begin to warm, beetle mortality decreases, and more beetles survive year to year. Furthermore, longer and warmer summers lead to increased development through their life stages, which starts a slow shift to overlapping life cycles. This is extremely dangerous, as population growth can increase rapidly and approach exponential rates. As populations rapidly grow, the beetles are more able to coordinate mass attacks that overwhelm the defense of many host trees within a forest. Pine beetle attacks begin when fully grown beetles emerge from trees in mid-summer. They first disperse in search of a suitable host tree, with beetles looking for trees under stress due to drought, age, overcrowding, or a volatile organic compound released by stressed trees. The first beetle to dig through a tree is a female called a pioneer beetle that lands on a tree and begins boring through the bark to test if the tree can be "colonized." If the pioneer beetle successfully enters the tree and determines that it is adequate, it releases aggregation pheromones, which are a chemical signal sent out to other beetles indicating that the tree is a suitable host, leading to a mass attack on the same tree. As more beetles begin colonizing the tree, the female beetles will release anti-aggregative pheromones, signaling that the host tree is becoming saturated, reducing further attraction of incoming beetles. At this point, large numbers of beetles arrive simultaneously, and each beetle bores its own tunnel into the bark. The tree attempts to defend itself with resin, which works against small numbers of beetles, but, during mass attacks, the resin defenses are overwhelmed, as the tree is unable to produce enough resin to stop all entry points. As a result, even healthy trees can be killed during outbreaks. Pine beetles carry a symbiotic blue-stain fungus, and when they enter the tree, they introduce this fungus into the phloem tissue



Model of pine beetle life cycle (Source: Caballero, 2022, GreenPal) beneath the bark. The fungus begins to spread, disrupting water transport within the tree and weakening the tree further. This creates a much more suitable environment for beetle larvae, speeding up tree mortality due to further water stress. Upon colonizing a tree, male and female beetles begin mating in the phloem. Next, females construct egg galleries under the bark, and lay eggs along the sides of these tunnels. Throughout this process, a single tree can host thousands of offspring. These eggs typically hatch within one to two weeks, depending on temperature, with warmer temperatures decreasing the amount of time it takes to hatch, ensuring the larvae emerge while the host tree and environmental conditions still support development. After the eggs hatch into larvae, they feed on the tree's inner tissues, further disrupting nutrient transport and damaging the tree's ability to survive even more. After a larva completes feeding, they stop eating and prepare for metamorphosis. They create small chambers under the bark, where they transform into pupae. During this process, the larval body breaks down, and the beetle starts to form wings, legs, and reproductive organs. Throughout this process, the beetle is inactive. Once their development is complete, new adults emerge from the tree in the following summer, and disperse to nearby trees to repeat the process. This coordinated attack allows mountain pine beetles to rapidly overwhelm tree defenses and expand infestations across entire forests.

A tree's ability to survive a pine beetle attack primarily depends on the strength of its natural defence systems, which can be significantly weakened under environmental stress. A tree's main defense against pine beetles is their resin, which is stored in resin ducts. When a beetle bores into the bark, the tree releases resin at the entry point, called a "pitch-out" response. When a tree does this, it floods the hole, pushing beetles out from their bark. Furthermore, as resin fills the hole, it can plug the tunnel that the beetle is trying to dig, preventing it from moving deeper into the tree. This often leads to the beetle being encased or blocked by resin. The resin also contains a toxic compound (terpenes), which can kill or repel beetles and blue-stain fungus. The production of resin is very dependent on the water pressure inside of the tree, but, during a drought, a lack of water will lower internal pressure, leading to slower resin flow and a tree failing to produce enough resin to repel attacks, so even a healthy tree becomes weak to pine beetle

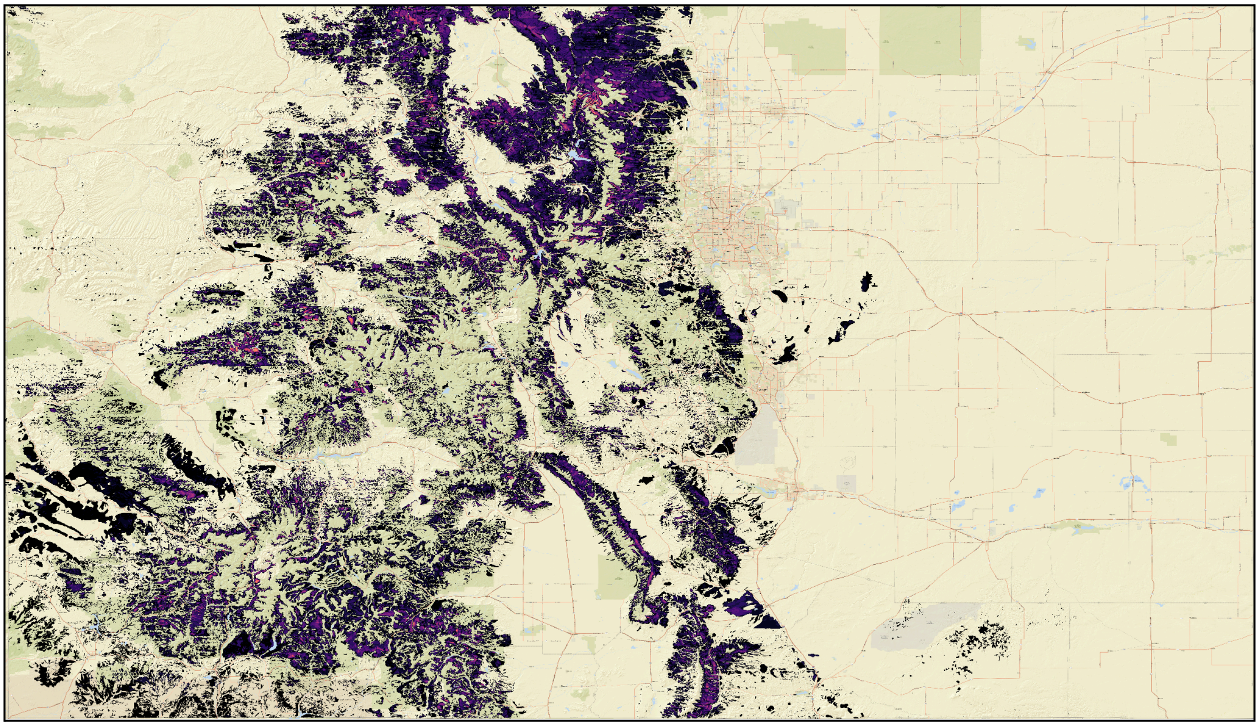
Resin "pitch-out" defense in a pine tree, showing a mountain pine beetle immobilized at the entry point (Photo by E. G. Vallery, USDA Forest Service, via Bugwood.org) infestation during a drought. Other factors of outbreak conditions are forest structure and age. Overcrowded forests increase stress on individual trees, with high tree density leading to competition for water, sunlight, and soil. This means that trees may already be operating under limited resources. Dense stands of trees also make it easier for beetles to spread, as trees being closer together means the beetles don't have to fly as far to find a new host tree. Pine beetles also prefer to attack older trees, as their reduced growth rates lead to less energy for defense, as well as thicker bark, which provides more habitat space for beetles, leading to larger beetle populations inside the tree. These factors rarely act in isolation, and typical outbreak conditions involve droughts, overcrowding, and aging forests. This leads to widespread weakening of trees across entire forests, meaning trees that would normally repel small beetle attacks become vulnerable to a mass attack.

Managing mountain pine beetle outbreaks involves coordination at the federal, state, and local levels. The primary focus is on forest health, wildfire reduction, and infestation control. At the federal level, the U.S. Forest Service (USFS) manages national forests, including the Roosevelt National Forest, where our data was collected from. Some of the main tactics used by the USFS are forest thinning, prescribed burns, and monitoring the forest. Forest thinning is where excess trees are removed to reduce overcrowding, which lowers competition for water and nutrients. Prescribed burns are carefully controlled fires with the intent to reduce accumulation of forest fuel, decrease wildfire risk, and promote stronger forest structure. Finally, aerial surveys are used to monitor forest health, allowing detection of mountain pine beetle outbreaks early and the ability to track the spread of infestations over time. At the state level, the Colorado State Forest Service (CSFS) coordinates forest health programs within Colorado, working with both landowners and federal agencies. The CSFS primarily focuses on mapping and reporting infestations, removing infested trees, helping private landowners manage infested trees, and reducing fire risk by removing dead trees. They are much more reactive than federal policy, and place emphasis on protecting communities and reducing immediate outbreak spread. At the local level, the Roosevelt National Forest management focuses on thinning trees in high-risk forest areas, removes beetle-killed trees along trails, roads, and other visited areas, and works with researchers to monitor outbreaks. Some of the biggest limitations of current approaches are that most restrictive measures are reactive, and it is quite difficult to stop spread once mass attacks begin. Furthermore, there is a limitation due to how large forests are, which makes it difficult to treat an entire population. Finally, climate change plays a major role in increasing pine beetle spread, but we don’t currently have solutions to warmer winters, as current policies focus on damage control, not climate driven population growth. Overall, existing policies across federal, state, and local levels focus primarily on mitigating the impacts of mountain pine beetle outbreaks, rather than eliminating the drivers of infestation.

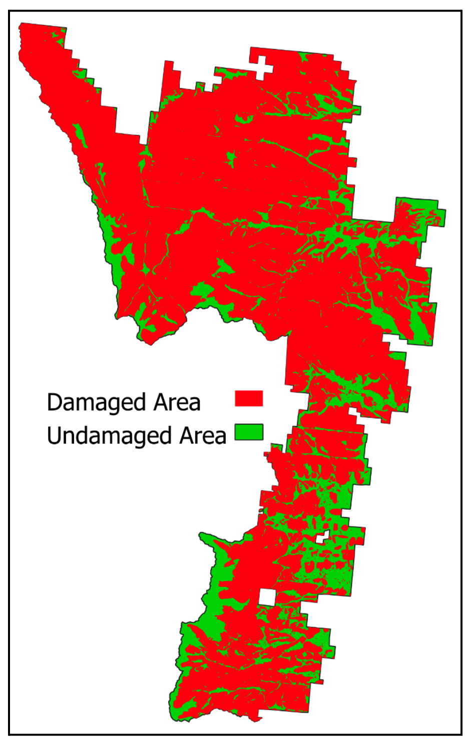
The Colorado State Forest Service (CSFS) serves as the primary stakeholder in this analysis due to its key role in managing mountain pine beetle impacts in Colorado. The CSFS is part of the Colorado State University system, which focuses on state and private forests, not federal forests. They work primarily with landowners, communities, and homeowners, while occasionally working with federal agencies. Their primary responsibilities are protecting forest health, keeping wildfires in check, and protecting the water supply. They monitor forest conditions across all of Colorado, identify pest outbreaks, and advise people on forest management strategies. They reduce wildfire risk by removing dead trees, which serve as fuel for forest fires, as well as encouraging fuel reduction practices to keep private land healthy. The CSFS is also very important when it comes to pine beetle outbreak prevention, as they map and track infestations, as well as assist landowners with removing infected trees and preventing spread to healthy trees. Because Colorado forests are a combination of federal, state, and private land, the CSFS is extremely important when managing beetle infestations. Beetle outbreaks often span across thousands of acres, and when the land they infest belongs to different people, the CSFS has to work with all groups to ensure all necessary precautions are taken everywhere, filling the gap between large scale federal land and local land. The primary goals of the CSFS are healthier and more resilient forests, reduced wildfire risk in Colorado, stronger economies through tourism and lumber, as well as increased access to public forests. Overall, the CSFS serves as a bridge between high-level forest management and community-owned land, aiming to reduce the impacts of mountain pine beetle outbreaks across all of Colorado.

03. Problem Description

Mountain pine beetle outbreaks have been a present and ongoing problem in Colorado for over a century, but it wasn't until the 1970s that their scope of damage became problematic enough to warrant action. Recent outbreaks have been far more severe, where the most recent survey published in 2017 shows that 3.4 million acres of trees have been killed by pine beetles. This is in addition to 70% of the 1.7 million acres of lodgepole pines across Colorado being damaged by pine beetles since the 70s. Seen in the map below is the millions of acres of pine beetle damage between the years 1997 and 2023 where a lighter color represents a higher and more intense concentration of



beetles. Further, as indicated by the data, nearly 860,000 acres were affected in Roosevelt national forest which is nearly 80% of the forest's total area. These staggering numbers show the scope of pine beetle's damage, and the particularly strong impact in Colorado, where millions of acres make up around a third of the total amount of acres impacted by pine beetles in the US.



Narrowing the scope further to our area of focus in Roosevelt national forest, we can see in figure on the right, the polygon in green representing the area of Roosevelt national forest outlined faintly. Layered on top in red is the area shown in the data to have been affected by pine beetles. Evidence of these widespread infestations over the years is seen in day to day life, with brown needled dead and dying pine trees standing by commercial buildings, in yards, and next to roads. Stands of beetle kill grows larger too in the mountains on the western half of Roosevelt national forest where areas visible from satellite imagery reveal recent beetle-caused mortality as large swaths of brownish trees faintly outlined among the dry grass of northern Colorado's already dry climate.

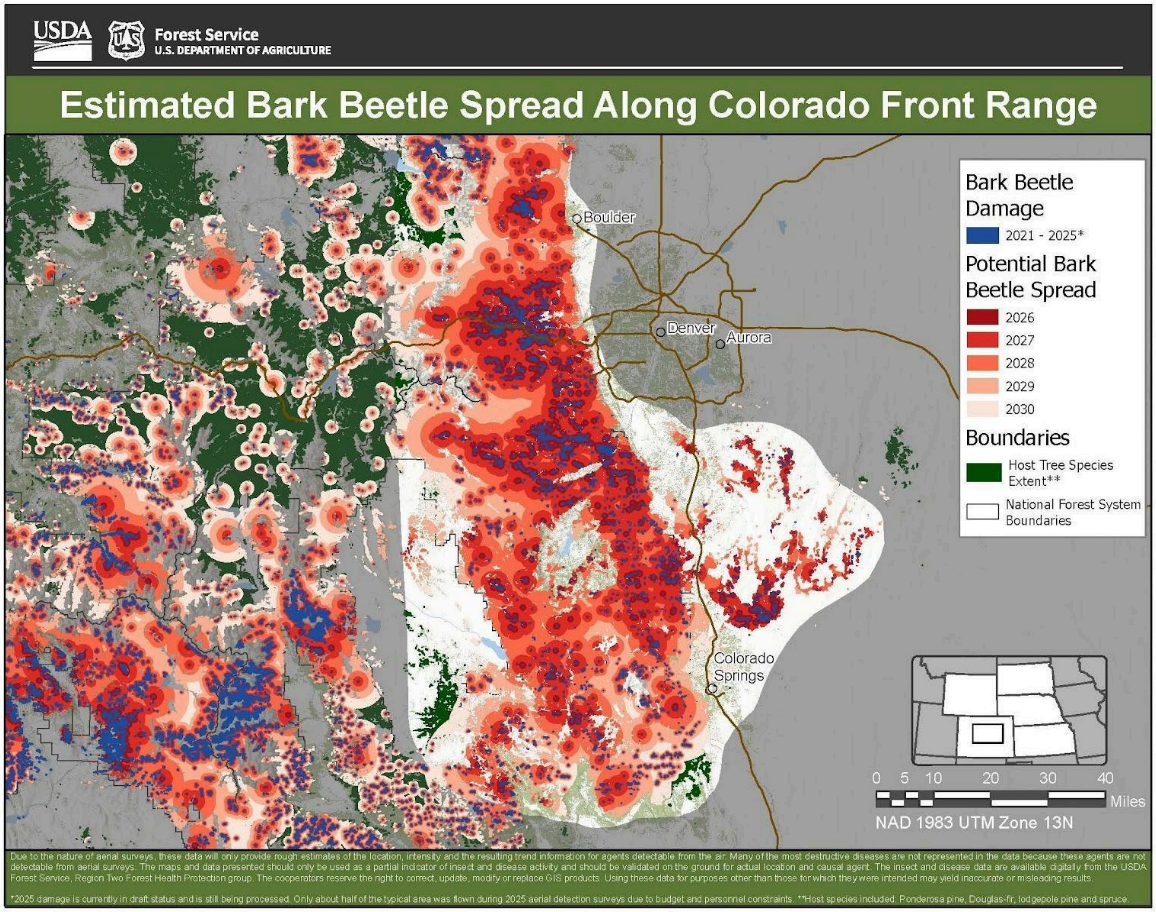
Tree mortality in the present day is highly damaging to local ecosystems but the potential dangers that stem from it are even more severe; namely forest fires. Colorado has in the past 25 years been in a drought most of that time, while only being out of a drought twice (Butzer, 2023). Along with present day being a quite severe drought, fire danger is extremely high and extrapolating the current trajectory of warmer drier winters leads to conditions that enable for hotter burning, much larger wildfires than have been seen in Colorado in quite some time.

Millions of Colorado residents are directly or indirectly impacted by beetle kill, but the task of managing them falls on relatively few organizations and individuals. The agency with the greatest stake in infestation management and prevention is the Colorado State Forest Service. This subsection of the Warner College of Natural Resources at Colorado State University has made a huge effort since its founding in 1955 to provide technical forestry assistance, wildfire mitigation expertise and outreach and education to help landowners and communities better assist in keeping forests healthy (Agency Profile). After their rapid rise of relevance and danger in the 70s and 80s, insect outbreaks and wildfire danger are now where most of CSFS's efforts and resources are directed.

The CSFS serves as an overarching, central sort of agency where they collaborate with communities, landowners, recreational organizations, and other agencies in order to coordinate efforts to improve forest health, reduce insect impacts, and promote healthy forests (Colorado State Forest Service, 2015). Their goal is to inform individuals and groups that are able to protect forests and to do this they must have as much knowledge and current data so that they can effectively and efficiently educate people on the action they should take. It is in the CSFS's interest to have access to as much research as possible with significant results that can be simplified and explained to a larger audience.

In being funded through the US Department of Agriculture it is imperative that (the cylinder remains unharmed) they are able to make the most of their budget and demonstrate the need to be funded. Thus in being Colorado's main agency for forest protection and health they have a large stake in how healthy Colorado's forests are.

Mountain pine beetle spread is the top of their priorities and thus it is very much in their interest to have more data and information on beetle spread and severity.



If there were no efforts to prevent pine beetle infestations throughout Colorado's forests there would be a mass dieoff magnitudes bigger than anything already seen in historic data. Recent models created by the US Forest Service show that of the most recent five years of beetle activity on the front range, pine beetles could affect the entirety of central Colorado forests by the year 2030 (adding citation at some point) The impacts of this would be felt by millions of Coloradeans as ecosystems would die leading to barren mountain sides and millions of dead trees by homes, trails, and other human utilized land. Aside from the aesthetic impact, all of this newly dried up wood and pine needles creates a perfect environment for hot and fast moving wildfires. This puts many people's lives in danger, as well as homes and so many more animals and their habitats.

As such it's CSFS's responsibility to put as much effort as they can into protecting forests from the worst possible case of beetle epidemic because if they don't there's simply just no forest left to protect. The fire danger caused by dying pine trees means that both of the Forest Service's main efforts of preventing pests and fires is served through protecting pine trees.

04. Solution Description

Although the actions the CSFS itself can take to address climatic conditions and worsening droughts are limited, there are several management options they can take to reduce the severity of Mountain Pine beetles' impacts.

One route would be to try and use direct controls, which reduce their population directly in the goal of keeping it below outbreak levels. There are several types of direct controls, some of the most effective of which are pesticides and trap trees. Pesticides can be sprayed on high-value trees to kill the pine beetles when they land on trees, or injected into them to kill the larvae before they emerge. One of the most common pesticides used is Monosodium Methanearsonate (MSMA), which is injected into the base of an infested tree, and is transported from the xylem into the phloem of a tree, and is then transported to and kills the beetle larvae. This is often used in trap trees, which use aggregative pheremones to first attract pine beetles to selected trees, and then are treated with MSMA to reduce the beetle population. One of the concerns of pesticide use is that if it's used on a widespread scale continuously, it can cause higher levels of inorganic arsenic in the soil, and can bioaccumulate in non-target species like woodpeckers (http://www.jstor.org.ezproxy.demo.willamette.edu/stable/40062119).

These can help control pine beetle populations until the conditions improve and trees are better able to resist them or winter mortality returns. They are primarily effective when pine beetle populations are low, and so would be best implemented when conditions favorable to pine beetle outbreaks are forecasted, but before pine beetle populations have started to rise. Once their population is high, they are much more unreliable. These methods also often require a high level of detection of infested tree and a large sustained effort lasting several years to be effective. These can be difficult to achieve, but if there are resources available to sustain a program, direct controls can be helpful in preventing pine beetle outbreaks (https://doi.org/10.3390/f5010103).

Another solution would be to utilize indirect controls to try and increase overall forest health, so the trees will better tolerate drought conditions and resist pine beetle attacks. One of the key techniques of this is tree thinning, which can be combined with landscape genomics. By preemptively cutting down certain trees in the goal of making a forest less dense, it can help give the remaining trees more space to grow, more nutrients, and more water, resulting in them being healthier and better able to resist drought conditions and be able to fight off pine beetles. It can also change the microclimate of forests, including wind patterns, which can disrupt the pheromone communication between beetles and reduce pine beetle infestations. One of the weaknesses of this is that it is not always reliably effective on its own. There are several examples of it failing to protect sections of forests, or even having higher mortality than unthinned stands. Tree thinning appears to be most effective when combined with intensive direct control treatments in surrounding areas (https://doi.org/10.3390/f5010103).

This can be combined with landscape genomics to better select which trees to be thinned. In landscape genomics, the genetic vulnerability of trees can be assessed to changes in temperature, precipitation, and the introduction of pathogens. As these are linked to genetic traits of the trees, if we thin out the trees that are most susceptible to these changes, and will be most vulnerable to mountain pine beetle infestation, the remaining pine trees will reproduce and create forests that are more resilient to drought and pine beetles (https://doi.org/10.3389/fpls.2018.00993). This is however, a rather new field, and could be difficult to implement across the wide scale needed to make a significant difference.

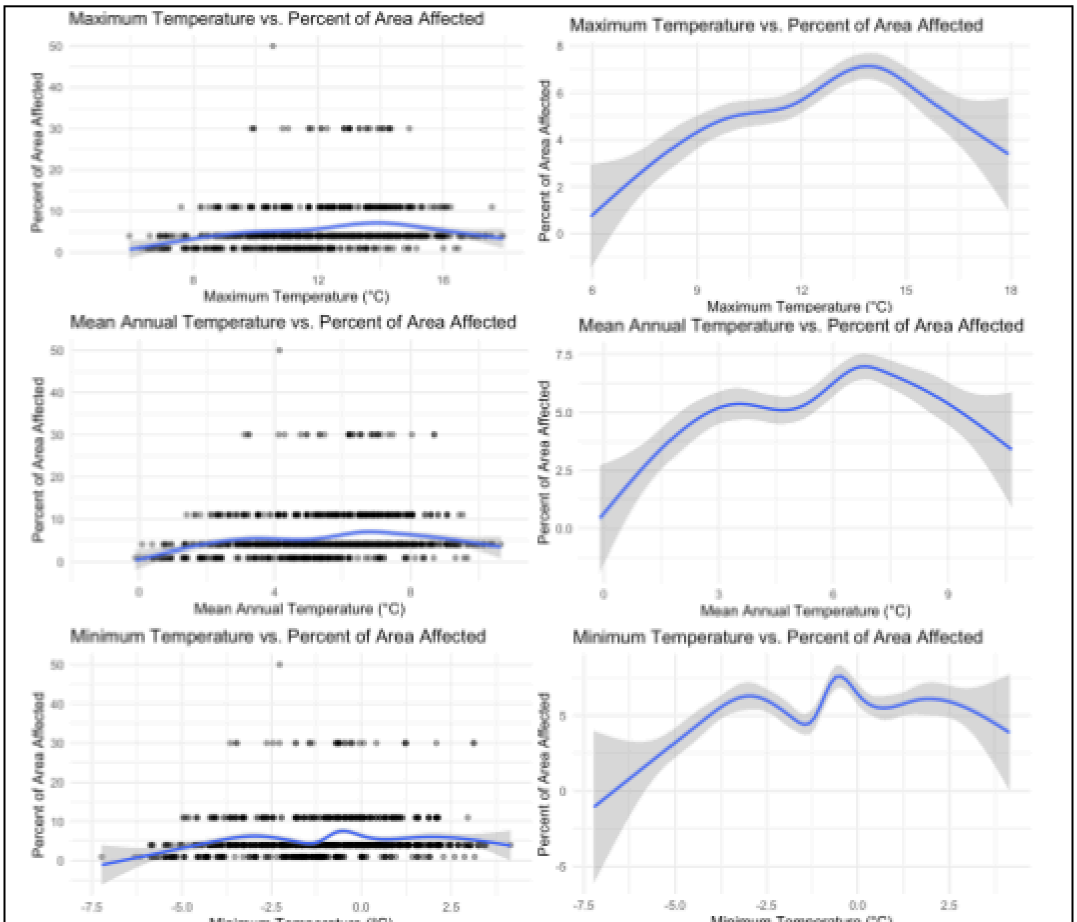
05. Recommendations (Needs Citations)

Pine beetle outbreaks pose a large, widespread, problem in forests throughout the United States and Canada. Therefore, controlling the spread of these pests is something that has no easy solution. It will take years of hard work and monitoring to know whether or not the recommendations presented in this paper will be of use. Countless ecologists have dedicated PhD studies, as well as years of research to this problem, and getting a solid understanding of these outbreaks is still very much in the works.

However, that does not mean that we cannot do anything. Combining all of the solutions we presented in the previous section we have devised a plan to help mitigate the spread of beetles in forests that may be prone to beetle attacks, or are already suffering from mass pine beetles. The most important thing to keep in mind with the presented problem is the huge scale that we are working on. Outbreaks can affect millions of trees per forest, and kill entire mountainsides. In order for any of our solutions to be feasible, they need to be done on a large scale for an extended period, which the Colorado State Forest Service (CSFS) cannot supply in their current numbers.

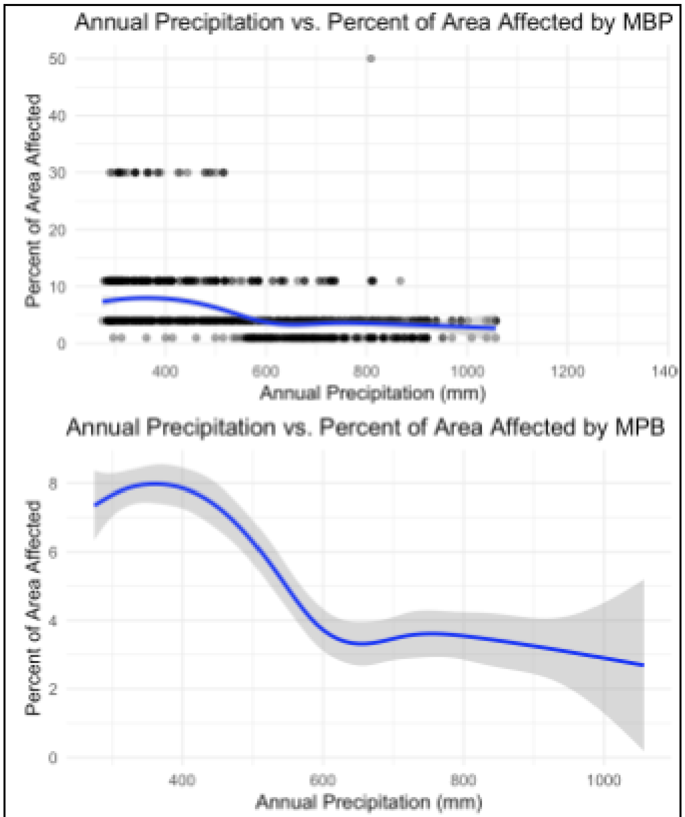
The CSFS is a rather small organization all things considered, being made up of only about 200 employees. To stop pine beetles from wiping out forests on large scales we will need more man power then that. Luckily, the CSFS puts most of their energy towards education about forest health. The Colorado forest service has partnered up with local wild fire brigades, as well as organizations like Boulder Outdoor Space and Mountain Parks (OSMP) to gain contributions to forest health.

We propose a campaign put on by the CSFS to request hands, either from pre-existing partnerships such as OSMP or by asking the public for volunteers. Then using existing literature the CSFS educated these people about pine beetles and the damage that they can do to forests. From there the field work begins. First, forest experts will go through a forest that they deem “at risk” for a mountain pine beetle outbreak. A forest could receive the label of “at risk” through a variety of reasons, including being an extremely dense forest, a forest that is already recovering from pine beetle damage, or a forest made up of mostly old and large trees. Next efforts to thin out the forest would begin. People would go through the selected forests and systematically select which trees are going to be cut down, while heartbreaking to be cutting down living trees is to work towards a healthier ecosystem as a whole. Of the trees selected to be removed several of them would be chosen to be trap trees. The trap trees would be dosed with the pine beetle’s aggregative pheromones to attract the beetles to particular trees. Once those trees become infested with the beetles they will be cut down and carefully removed from the forest, taking the pine beetles with them. Due to how dense many forests are, not



The figures on the left show the raw temperature data, whereas the figures on the right zoom into the linear regression where we can see the patterns emerge. The y axis is percent affected and the x axis for all three graphs is temperature. As seen, the maximum and mean temperature figures show a similar pattern both having the highest percent of area affected by pine beetles in the middle. The minimum temperature graphs show really interesting results however, illustrating that the pine beetles like milder/warmer winters. **[Cite Data Set in caption]** every tree to be removed would be a dedicated trap tree. This means that the trees being removed simply to get the forest to a healthy density could be sent to lumber yards and used to make wooden products. This would serve another purpose of limiting deforestation efforts due to lumber production.

Logically the next question in this line of thought is “how do we know which forests to target” which has an easy answer. Research suggests that pine beetles prefer climates that have mild/warmer winters. This is due to the fact that Colorado can get very cold and these beetles struggle to survive in extremely frigid temperatures. From our findings, pine beetles also seem to dislike extreme heat, having a sweet spot around the moderate temperatures. These temperature preferences can be seen in the given temperature plots.



The top figure shows the raw data and the second figure zooms in to show the linear regression pattern. We can see a sharp drop in the area affected as precipitation increases. **[Cite Data Set in caption]**

Another way that we can choose which forests to target with pine beetle mitigation plans is by looking at precipitation. Our findings, as well as the existing literature, find that pine beetles prefer dryer climates and droughts, over lots of precipitation. This means that
We can look at annual rainfall patterns to find forests in droughted areas and try to protect those forests from possible attacks.

A solution that was mentioned in the solutions section but not mentioned in our recommended plan of attack is the use of insecticides and pesticides. There have been some insecticides developed that specifically target m ountain pine beetles. The issue with using insecticides is that they can only be used on a small scale (single trees at a time), they are also made up of chemicals that can be harmful to other bugs that may be helpful to the ecosystem or the ecosystem as a whole. In small amounts, mountain pine beetles can also be beneficial to forests, due to their methods of clearing out dead and dying trees, thus, mass use of insecticides could kill more beetles than we wish to kill, leading to new problems in forest health. However, with all of that said, there are scenarios where insecticides could be beneficial when used in controlled amounts. One of these scenarios could be a drought climate where prescribed burns are difficult due to risk of losing control in dry places with lots of fuel, and lack of rainfall means that less of the chemicals will be washed away into water supplies.

06. Conclusions

Restate points mentioned in executive summary.

07. References

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